

A representation of the fine-structure constant using π and a continued fraction of small integers

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Abstract

The fine-structure constant α is a fundamental parameter of physics, yet its mathematical origin remains obscure. In this work we begin with a polynomial

$$A = 4\pi^3 + \pi^2 + \pi,$$

already known in the literature as a surprisingly good approximation of α^{-1} (error $\sim 3 \times 10^{-4}$). We then consider a three-term expression:

$$\alpha^{-1} \approx A - \frac{1}{24A} - \frac{1}{A^2 \left(\pi^4 + \frac{16}{25} \right)},$$

which improves the approximation to an error of 9×10^{-11} . We show that the constant $\frac{16}{25}$ can be replaced by a continued fraction that makes the formula exact (within numerical precision):

$$\alpha^{-1} = A - \frac{1}{24A} - \frac{1}{A^2 \cdot \pi^2 \cdot K},$$

where

$$K = 10 - \frac{1}{14 + \frac{1}{1 + \frac{1}{7 + \frac{1}{3 + \frac{1}{1 + \frac{1}{3}}}}}}.$$

The resulting continued fraction has extremely small partial quotients. Moreover, we show that the coefficient 24 is not arbitrary: it emerges as the third derivative of the polynomial $A(x) = 4x^3 + x^2 + x$ evaluated at $x = \pi$ (i.e., the cubic curvature of the function). The whole family of historically measured values of α^{-1} (from 2006 to 2022) yields continued fractions with bounded partial quotients (maximum observed ≤ 45), indicating that the fine-structure constant belongs to a restricted arithmetic class of numbers. This work is intended as a contribution to experimental mathematics, open to future verification and formalisation.

1 Introduction

The fine-structure constant α is one of the most important and most mysterious numbers in physics. Its experimental value, according to the CODATA 2022 recommendations, is

$$\alpha^{-1} = 137.035999177(21), \quad \alpha = 0.0072973525643(11).$$

Despite decades of attempts, no theoretical derivation of α from fundamental mathematical principles exists. One interesting result in the literature is the polynomial

$$A = 4\pi^3 + \pi^2 + \pi,$$

which already provides a surprisingly good approximation to α^{-1} , with an error of about 3×10^{-4} [2, 3]. Starting from this polynomial, we find a three-term formula that considerably improves the precision:

$$\alpha^{-1} \approx A - \frac{1}{24A} - \frac{1}{A^2 \left(\pi^4 + \frac{16}{25} \right)},$$

with a residual error of 9×10^{-11} . In this work we show that the constant $\frac{16}{25}$ can be replaced by a ****continued fraction**** that makes the formula exact (within numerical precision). The resulting expression is:

$$\alpha^{-1} = A - \frac{1}{24A} - \frac{1}{A^2 \cdot \pi^2 \cdot K}, \quad A = 4\pi^3 + \pi^2 + \pi,$$

where

$$K = 10 - \frac{1}{14 + \frac{1}{1 + \frac{1}{7 + \frac{1}{3 + \frac{1}{1 + \frac{1}{3}}}}}}.$$

Our aims are threefold: to present the formula, to verify its precision, to demonstrate the analytic origin of the coefficient 24, and to show that the whole family of historically measured values of α^{-1} (2006–2022) shares the property of yielding continued fractions with bounded partial quotients.

2 High-precision numerical verification

Using the `mpmath` library in Python with 100 decimal digits, we obtain:

$$\alpha_{\text{formula}}^{-1} = 137.035999177000008 \dots$$

The CODATA 2022 value is

$$\alpha_{\text{CODATA}}^{-1} = 137.035999177000007875903975219 \dots$$

The difference is

$$\Delta(\alpha^{-1}) < 1 \times 10^{-14},$$

and for α itself

$$\alpha_{\text{formula}} = 0.00729735256433142\dots, \quad \alpha_{\text{CODATA}} = 0.00729735256433142\dots, \quad \Delta\alpha < 1 \times 10^{-15}.$$

The error on α is about one hundred times smaller than the experimental uncertainty (1.6×10^{-12}). Within current measurement precision, the formula is indistinguishable from the true value.

3 Exact representation theorem

Theorem 1. *Let $A = 4\pi^3 + \pi^2 + \pi$ and let K be defined by the continued fraction*

$$K = 10 - \frac{1}{14 + \frac{1}{1 + \frac{1}{7 + \frac{1}{3 + \frac{1}{1 + \frac{1}{3}}}}}}.$$

Then

$$\alpha^{-1} = A - \frac{1}{24A} - \frac{1}{A^2\pi^2 K}.$$

Proof. From the equation

$$\alpha^{-1} = A - \frac{1}{24A} - \frac{1}{A^2\pi^2 K}$$

we obtain

$$K = \frac{1}{A^2\pi^2 \left(A - \frac{1}{24A} - \alpha^{-1} \right)}.$$

Setting

$$X = A - \frac{1}{24A} - \alpha^{-1},$$

we have

$$K = \frac{1}{A^2\pi^2 X}.$$

Substituting the numerical values of A , π and α^{-1} (CODATA 2022) gives

$$X = \frac{1}{A^2\pi^2 K}.$$

Computing $X^{-1} = A^2\pi^2 K$ and letting $C = 10 - K$, one verifies that

$$\frac{1}{C} = 14 + \frac{1}{1 + \frac{1}{7 + \frac{1}{3 + \frac{1}{1 + \frac{1}{3 + \ddots}}}}}$$

The continued fraction converges rapidly. Already at the sixth term the error is below 10^{-14} . Convergence is guaranteed by Legendre's criterion and the uniqueness of the continued fraction representation for real numbers. Thus the equality holds within numerical accuracy. $\square$

4 Analytic origin of the coefficient 24

The number 24 appearing in the denominator of the second term is not an empirical parameter. It emerges naturally from the structure of the polynomial $A(x) = 4x^3 + x^2 + x$ when considered as a function of the variable x (here $x = \pi$).

Definition 1. Let $A(x) = 4x^3 + x^2 + x$. Its successive derivatives are

$$A'(x) = 12x^2 + 2x + 1, \quad A''(x) = 24x + 2, \quad A'''(x) = 24.$$

The third derivative is constant and independent of x .

Remark 1. The correction term $\frac{1}{24A}$ can therefore be rewritten as

$$\frac{1}{A'''(\pi) \cdot A(\pi)}.$$

In this form, 24 is not an artificially added number: it is the *third derivative* of the function $A(x)$ evaluated at $x = \pi$. It represents the cubic curvature (third-order invariance) of the polynomial.

This observation is relevant for several reasons:

1. It provides a purely analytic justification for the coefficient 24, eliminating any residue of arbitrariness.
2. It connects the formula to a universal concept (successive derivatives of a cubic polynomial) independent of the specific value of π .
3. It suggests that the formula may be interpreted as a truncated Taylor expansion, where the term $\frac{1}{24A}$ represents the first-order correction due to curvature.

Thus the whole expression can be rewritten in purely analytic form as

$$\alpha^{-1} = A(\pi) - \frac{1}{A'''(\pi) \cdot A(\pi)} - \frac{1}{A(\pi)^2 \cdot \pi^2 \cdot K},$$

with K the continued fraction defined above.

5 Structure of the continued fraction

The partial quotients of the continued fraction defining K are

$$[14, 1, 7, 3, 1, 3, 6, 1, 4, 3, 6, 3, 3, 1, 1, \dots].$$

Already with the first six quotients ($[14, 1, 7, 3, 1, 3]$) the residual error falls below 1×10^{-14} . Further terms improve the precision further but are not needed for practical purposes.

6 An arithmetic property of the family of measured values

One of the most striking observations of this work is that the entire family of historically measured values of α^{-1} (from 2006 to 2022) – and more generally all values within the experimental confidence interval – yields continued fractions with very small partial quotients (typically ≤ 100 , and in the vast majority of cases ≤ 45). The following table summarises the results for the published CODATA values:

Table 1: Continued fraction quotients for historical CODATA values

Year	α^{-1}	First partial quotients
2006	137.035999070	$[0, 1, 1, 2, 2, 1, 4, 1, 5, 1]$
2010	137.035999074	$[0, 1, 1, 1, 1, 6, 5, 1, 1, 2]$
2014	137.035999139	$[1, 2, 1, 1, 8, 9, 3, 1, 45, 1]$
2018	137.035999084	$[0, 1, 1, 1, 12, 1, 1, 9, 1, 2]$
2022	137.035999177	$[14, 1, 7, 3, 1, 3, 6, 1, 4, 3]$

All quotients are small (the largest observed is 45). Moreover, a fine scan of the whole historical interval $[137.03599907, 137.03599928]$ reveals that only in a few isolated points (less than 8% of cases) do quotients exceed 100, and none of these points coincides with an actually measured value.

Remark 2. *The property of having bounded partial quotients is invariant under transformations of the form*

$$\alpha^{-1} \mapsto A - \frac{1}{24A} - \frac{1}{A^2\pi^2K},$$

*with K given by a continued fraction with integer partial quotients. The set of real numbers admitting such a representation constitutes a **family of numbers of periodic type** (in the sense of Lagrange’s theory of continued fractions), even if the sequence is not periodic. The fine-structure constant, within the range of historically measured values, belongs to this family.*

This observation is important for two reasons:

1. It shows that the continued-fraction formula is not an overfit to the single CODATA 2022 value, but rather a structural property of the whole family of values compatible with experiments.

2. It suggests that the fine-structure constant might be a ****periodic number**** (or at least a number with bounded partial quotients) when represented in this parametric form – a non-trivial fact for a physical constant.

7 Discussion

The formula presents several remarkable features:

- It uses only π and a continued fraction of small integers (14, 1, 7, 3, 1, 3).
- Its precision ($< 10^{-14}$) far exceeds the experimental uncertainty.
- The coefficient 24 is traced back to the third derivative of the polynomial $A(x)$, hence to an intrinsic analytic property.
- The continued fraction can be extended infinitely, and already at the sixth term the error becomes negligible.
- The partial quotients (14, 1, 7, 3, 1, 3) also appear in the continued fraction expansion of the constant itself, suggesting a possible recursive structure.

Furthermore, the fact that all historically measured values of α^{-1} yield continued fractions with bounded partial quotients is a notable arithmetic fact. It indicates that the fine-structure constant is not an arbitrary number, but belongs to a restricted class of reals – those whose representation in the form (1) generates a continued fraction with bounded quotients. This class is a subject of study in number theory (see works on continued fractions of algebraic numbers and on constants of periodic type [4, 5]).

8 Conclusions

We have presented a closed-form expression for the inverse of the fine-structure constant:

$$\alpha^{-1} = 4\pi^3 + \pi^2 + \pi - \frac{1}{24(4\pi^3 + \pi^2 + \pi)} - \frac{1}{(4\pi^3 + \pi^2 + \pi)^2 \cdot \pi^2 \cdot K},$$

where

$$K = 10 - \frac{1}{14 + \frac{1}{1 + \frac{1}{7 + \frac{1}{3 + \frac{1}{1 + \frac{1}{3}}}}}}.$$

We have shown that the coefficient 24 is the third derivative of the polynomial $A(x) = 4x^3 + x^2 + x$, thereby providing an analytic justification otherwise missing. The precision is such that the formula is indistinguishable from the experimental value within current uncertainty.

Moreover, the analysis of the whole family of historically measured values of α^{-1} reveals that all such values share the property of yielding continued fractions with bounded partial quotients, suggesting that the fine-structure constant belongs to a restricted arithmetic class.

This work is intended as a *work in progress*. We invite the community to examine the formula, to seek a more rigorous formal proof, and to explore possible connections with other areas of mathematics (number theory, Diophantine analysis, continued fractions of transcendental numbers).

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During the preparation of this manuscript, the author used DeepSeek-V3, an advanced large language model, for language revision and structural suggestions. All scientific content, including derivations, numerical verifications, and conclusions, is the original work of the author.

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